

Measuring snow microstructure and hardness using a high resolution penetrometer

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Abstract

Using a high resolution snow penetrometer we are able to measure snow penetration resistance and snow meso- and microstructure. The variation of the signal is characteristic for different snow types. The penetrometer can be used in the field as well in the laboratory. The range of snow types which can be tested extends from light new snow (50 kg m^{-3}) to very dense snow occurring on ski race tracks (500 kg m^{-3}). The displacement resolution of the penetrometer is better than one mm to detect significant changes in resistance, the force signal is measured every 0.004 mm. The constant penetration velocity can be varied between 6 and 20 mm s^{-1} . A classification is developed to extract textural information of snow from the force signal. The classification is based on experiments with artificially metamorphosed snow and undisturbed snow from the Alps and Alaska. Two weak layers were identified and compared to surface sections. The new instrument makes the measurement of mechanical and stratigraphic features of a snowpack a more objective and easy task than with other methods. The high displacement resolution promises also a better identification of weak layers. © 1999 Elsevier Science B.V. All rights reserved.

Keywords: Snow hardness; Metamorphism; Snow strength; Snow stratigraphy; Texture

1. Introduction

The texture of snow (Arons and Colbeck, 1995) and its associated mechanical and physical parameters are of key importance to most scientific questions dealing with snow. Examples are mechanical properties of ski tracks, avalanche formation and availability of food to animals. However, a quantitative and rapid method for measuring the texture of snow is not currently available. At present, the meth-

ods to classify snow rely heavily on the experience of the user. In addition, the discrimination of layers is often very difficult and not without personal bias. Laboratory methods (surface or translucent sections, photography of single grains) give quantitative information. However, they are time consuming and the interpretation of the snow properties is often very difficult to do because stereological methods to characterize the mechanical properties of low density materials with complicated shapes are not well developed (Lesaffre et al., 1998). Sieving gives at least a quantitative grain mass distribution, but loses all information about bonding properties. Schneebeli and

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Johnson (1998) developed a high resolution penetrometer to measure the stratigraphy and penetration resistance. This penetrometer has a high resolution in penetration resistance, but interpretation methods to obtain structural information were not developed. Schneebeli and Johnson (1998) also reviewed previously developed penetrometer and indicated their shortcomings. In this paper we demonstrate that our high resolution penetrometer, with a higher measure-

ment frequency, can delineate different snow texture types. This should enable us to measure snow texture and its mechanical properties much more objectively and rapidly.

2. Construction

The penetrometer (called SnowMicroPen) consists of a small flared measuring tip, 5-mm diameter with



Fig. 1. Field version of the SnowMicroPen. The motor housing (below the hand of the operator, the drive cone, the mounting rods, the electronics box and the computer are shown (photo: A. Roth, SLF).

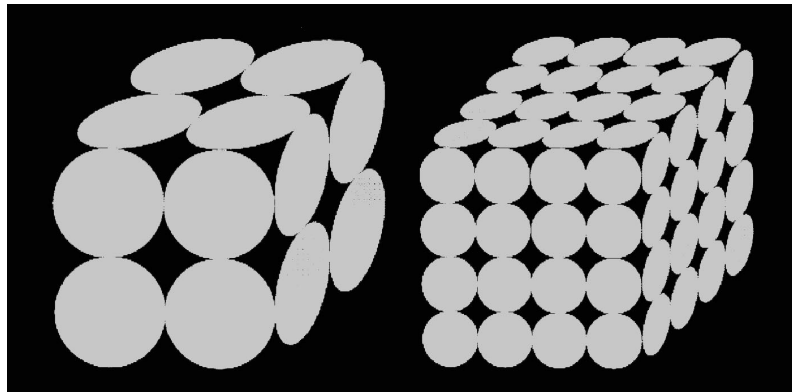


Fig. 2. A cube consisting of large grains has fewer contact points than a cube of the same dimension consisting of small grains. Both cubes have the same volumetric density. The TI of the left configuration is 3.8, while the cube on the right has a TI of 1.9.

a 60° tip angle. This tip is connected to a high resolution force transducer which is housed in a drive cone and rod of 16-mm diameter. The drive rod is geared and driven by a constant speed rotary

motor (Johnson and Schneebeli, 1998). The force transducer has a range of 0–500 N and a resolution of 0.01 N. The penetration velocity is 20 mm s^{-1} , but can be varied between about 6 and 20 mm s^{-1} .

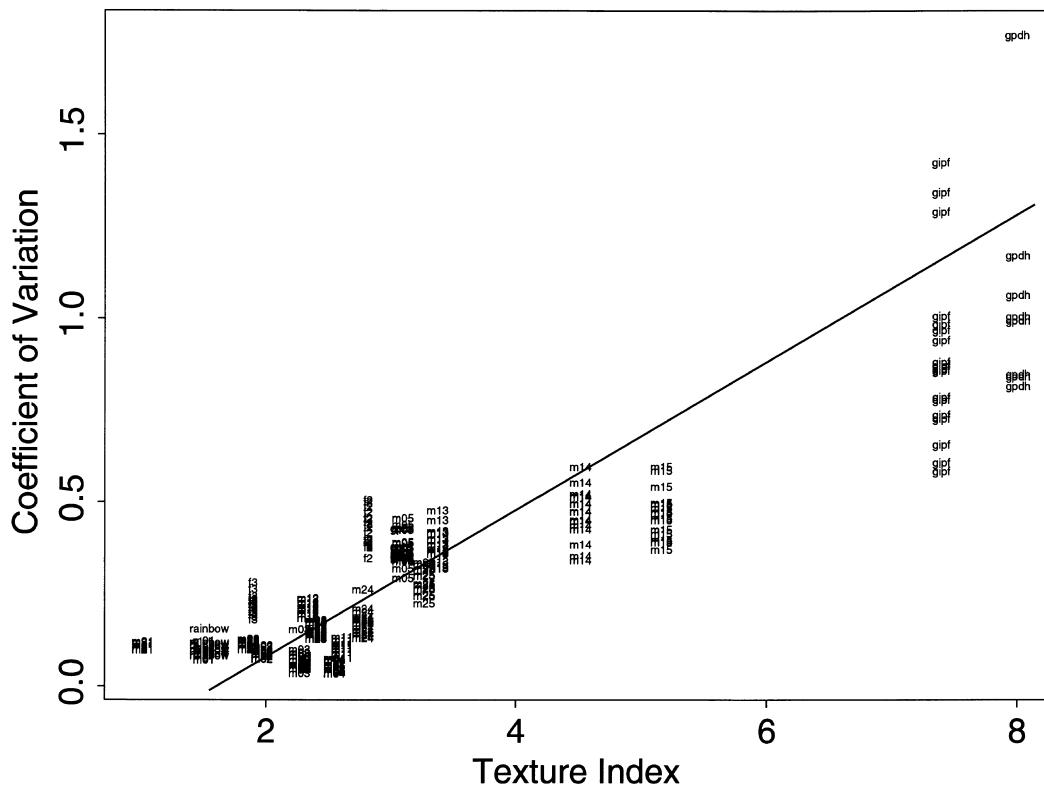


Fig. 3. Relation between TI and coefficient of variation. The best linear fit is also shown. The symbols stand for different snow types. Each measurement is calculated over a length of 4 mm.

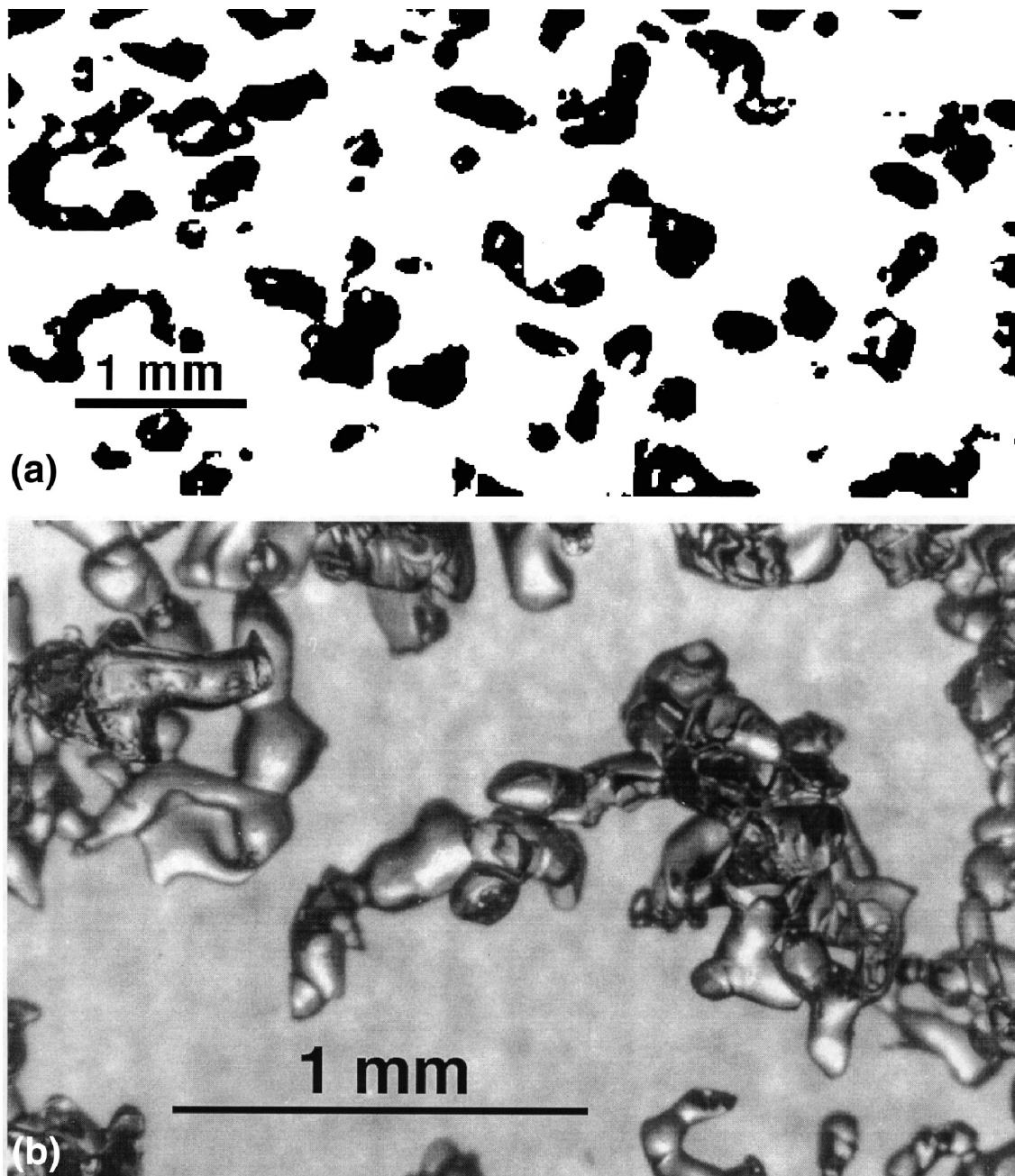


Fig. 4. (a) Surface section of about 3-month-old Alaskan equitemperature metamorphosed snow. On this and the following surface sections, ice is represented by black and air by white. (b) Grain photography (photo: M. Sturm) of 3-month-old equitemperature metamorphosed snow.

A force measurement is acquired every 4 μm , independent of penetration velocity. At the most often used velocity of 20 mm s^{-1} the sampling frequency is 5000 Hz.

The electronics are housed in a small box. Data transfer is via the parallel interface to a notebook computer. The data acquisition software runs on the MS-DOS operating system. The motor and the drive cone (length 1.7 m) are mounted on two rods with an included 60° angle (Fig. 1). The total weight of probe and electronic box is 8.5 kg. Measurements are always taken perpendicular to the slope. This is different from the usual gravity oriented measurement practice using the classical rammsonde. Measurements perpendicular to the slope show less error at the interface going from a weak layer to a hard or crusty layer. The reaction force is exerted by leaning on the motor housing. Experience shows that snow layers with a hardness up to knife (Colbeck et al., 1990) can be easily measured. Even harder snows can be measured using a fixed frame (laboratory version) or a specially designed sledge, which can be fixed with snow screws.

One high resolution force recording of 1.5-m depth has 375,000 readings. Depending on the format of storage, this results in a file size between 750 KB and about 2.5 MB. To minimize the size of the files the numbers are stored in binary format. A custom program creates graphics from the data files and can export the files in ASCII format.

3. Definition of a textural index

It is well known that the mechanical properties of snow are not only dependent on density, but also much influenced by its texture. Texture is often described in terms of grain shape. During the evaluation of the penetration measurements we found that grain shape is not ideally suited to parameterize the structural changes occurring in snow. We found that the ratio of mean grain size [μm], determined by sieving and calculated after Friedman and Sanders (1983) divided by the density of snow [kg m^{-3}]

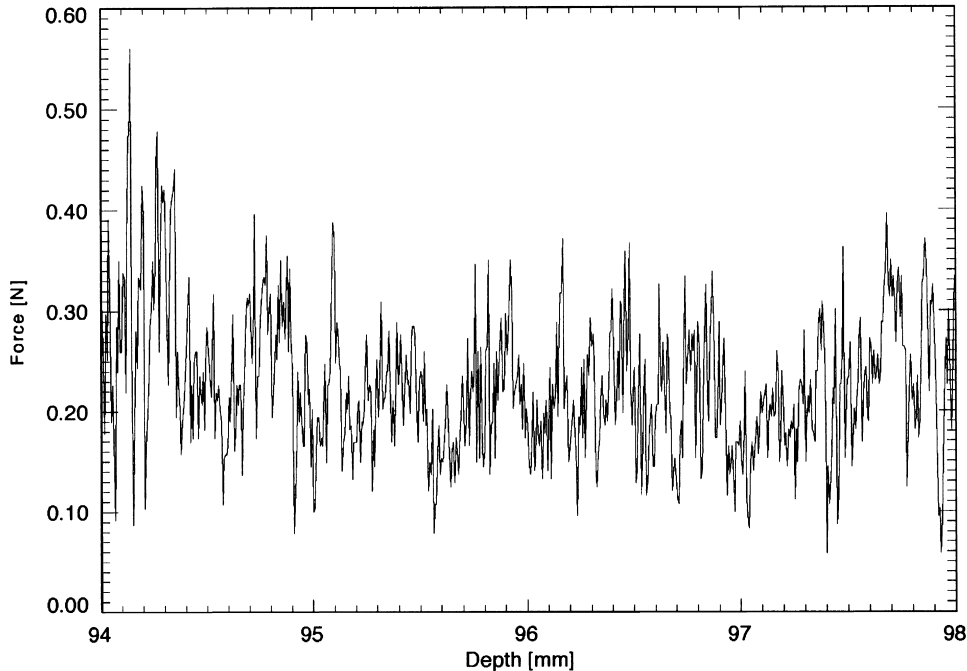


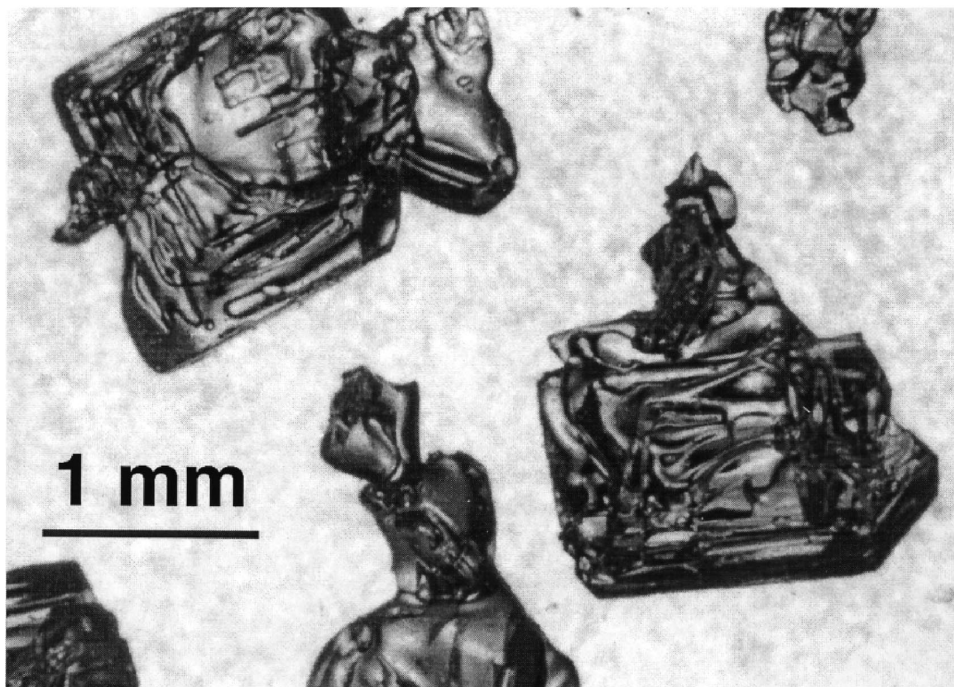
Fig. 5. Force–distance signal of the penetration measurement made in equitemperature metamorphosed fine grained snow shown in Fig. 4.

gives us a good measure. The calculation of the texture index (TI) is based on the idea that smaller spherical structural elements have a decreasing ratio

between mean grain size and volumetric density, given the same density (see Fig. 2). The TI is also a direct index of the volume density of grain contacts.



(a)



(b)

Fig. 6. (a) Surface section of Alaskan depth hoar snow. (b) Grain photography (photo: M. Sturm) of Alaskan depth hoar snow.

4. Correlation of the textural index to measurements

To correlate the TI to the force signal of the SnowMicroPen we measured 23 different snow samples (Fig. 3). These snow samples cover all five major snow classes (Colbeck et al., 1990). The measured texture index, TI_m , was determined by measuring density and mean grain size by sieving. Each penetration measurement had a length of about 0.2 m. The penetration force measurements were split in sections of 1000 data points, corresponding to 4 mm of penetrated snow and the coefficient of variation (CV) (standard deviation divided by the mean) was calculated on each segment. A segmentation length of less than 4 mm showed a strong fluctuation of the resulting CV. A linear regression model resulted in the following coefficients:

$$TI = 1.45 + 5.72CV \quad (1)$$

The coefficients are highly significant ($p < 0.001$) and the coefficient of correlation r is 0.89. Based on

this result, the TI can be calculated easily from penetration measurements.

5. Case studies

The SnowMicroPen has been tested on a wide range of snows, from light new snow (50 kg m^{-3}) to very dense snow occurring on ski race tracks (500 kg m^{-3}). We will describe in this section several case studies. The texture resolving capability is demonstrated first on two snows from Fairbanks, AK. The snow shown in Fig. 4 was deposited on a wooden table in an open area. Only small and reversing temperature gradients (TG) developed due to the access of air to the underside of the table. The density of the snow is 229 kg m^{-3} , and the mean grain size, determined by sieving, is 0.7 mm. A small part of the penetration measurement (4 mm of a total of 120 mm of the sample) is shown in Fig. 5. The signal varies rapidly between peaks and valleys. A lower frequency component can be observed with a range of about 0.2 mm, as well as a higher frequency component with a range of about 0.01

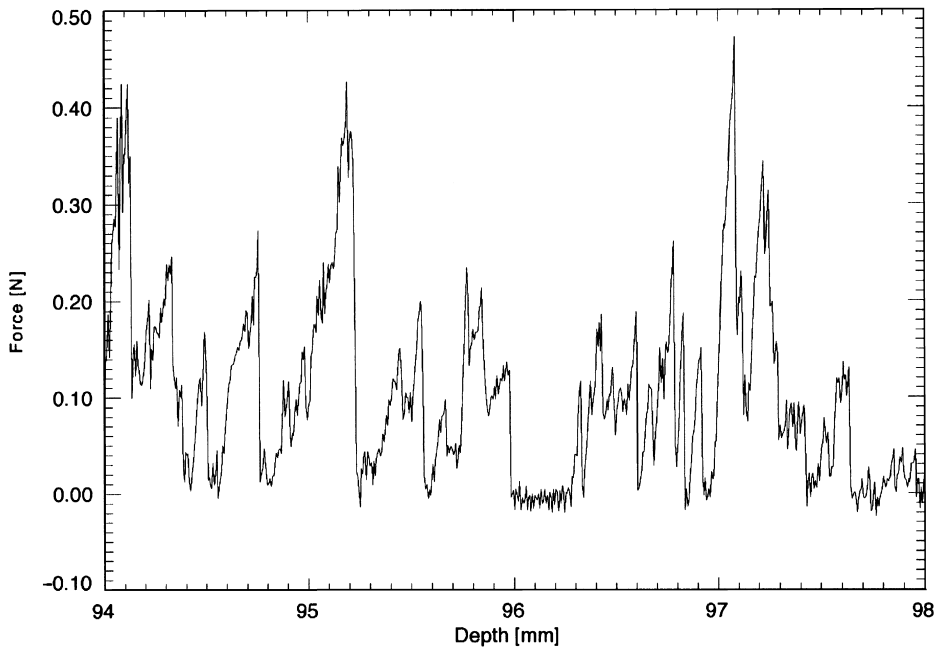


Fig. 7. Force–distance signal of the penetration measurement made in depth hoar shown in Fig. 6.

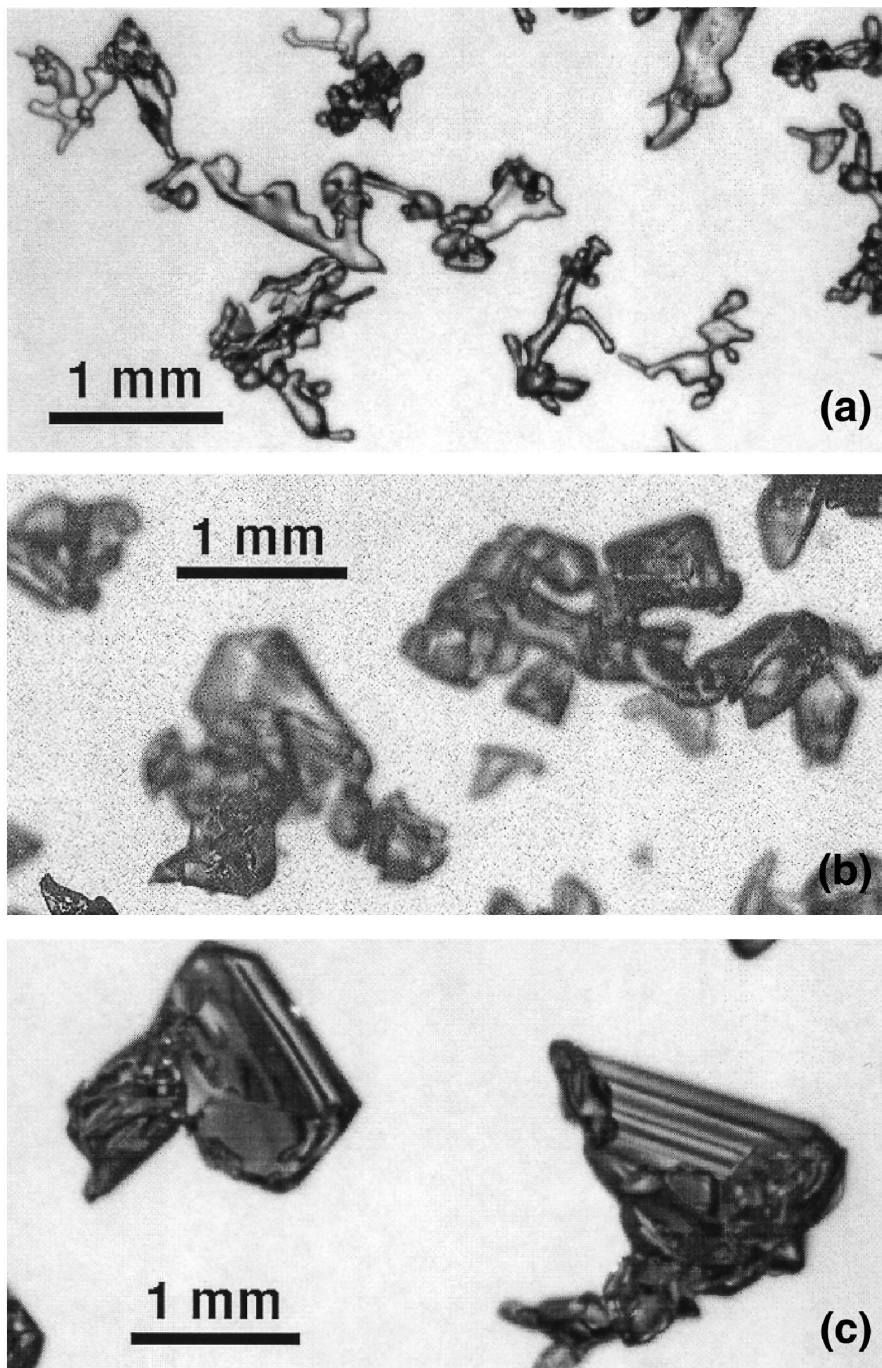


Fig. 8. (a) TG experiment at start. (b) TG experiment after 1 week. (c) TG experiment after 2 weeks. (d) TG experiment after 3 weeks. (e) TG experiment after 4 weeks.

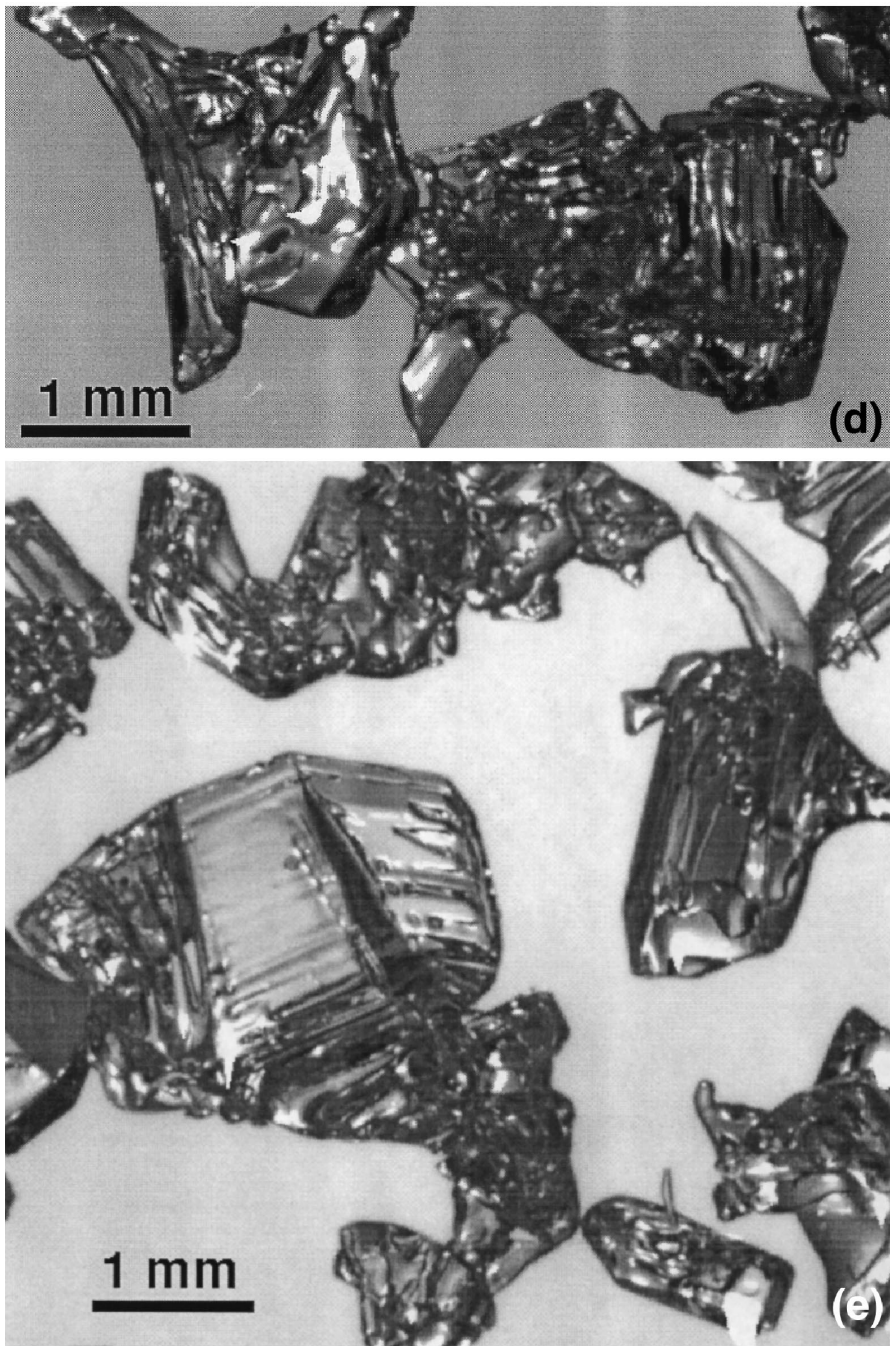


Fig. 8 (continued).

mm. The higher frequency range is at the detection limit of the displacement resolution of the penetrometer.

The snow shown in Fig. 6 is from the same location as the equitemperature (ET) metamorphosed snow, but from the ground. Large TGs across the

Table 1

Parameters of the snow metamorphosed under controlled conditions: day of sampling after TG applied, mean grain size, determined by sieving and density

Day	0	7	14	23	30
Mean grain size [mm]	0.46	0.52	0.76	1.1	1.15
Density [kg m^{-3}]	178	225	227	245	224

snow produced a mature depth hoar. The density is 238 kg m^{-3} , only slightly higher than the density of the ET metamorphosed snow. The mean grain size is 1.9 mm. The penetration measurement (Fig. 7) is very different from Fig. 5. It shows a distinct zigzag pattern, consisting of a rising slope in the direction of the penetrometer movement, and a very rapid fall. This pattern shows a typical separation distance of 0.2–0.4 mm. The mean penetration force is 0.16 N in the case of the ET snow, and half of that, 0.08 N in the case of the depth hoar. The peak resistance is 0.54 N for the ET snow, and 0.78 N for the depth hoar (statistics from complete sample, penetration length 120 mm). This result can be interpreted either with a micromechanical model (Johnson and Schneebeli, 1999) or by a statistical approach, as will be used here.

Table 2

The TI_m (calculated from mean grain size and density), the CV from the penetrometer resistance, and TI calculated based on the regression (Eq. (1))

Day	0	7	14	23	30
$\text{TI}_m [10^{-3} \text{ m}^4 \text{ kg}^{-1}]$	2.6	2.3	3.4	4.5	5.2
CV [%]	0.1	0.22	0.4	0.43	0.47
TI [$10^{-3} \text{ m}^4 \text{ kg}^{-1}$]	2.0	2.7	3.7	3.9	4.1

Our interpretation of the natural snow penetrometer measurements were confirmed using controlled TG experiments. New snow was sieved homogeneously into a well insulated box with a heating plate at the bottom. The bottom was kept at -2°C , while the top was at -10°C . 20 cm of snow were in the box, resulting in a TG of 40°C m^{-1} . The changes in snow crystal form and size are shown in Fig. 8. The changes in snow density and mean grain size are shown in Table 1. The penetration resistance of these different snow types is shown in Fig. 9. The median force and its variability decrease until week 2. At this time most of the grains are faceted, and the CV (standard deviation/mean) is steadily increasing (Table 2). The snow shows its minimal resistance, and we interpret this as the period when most origi-

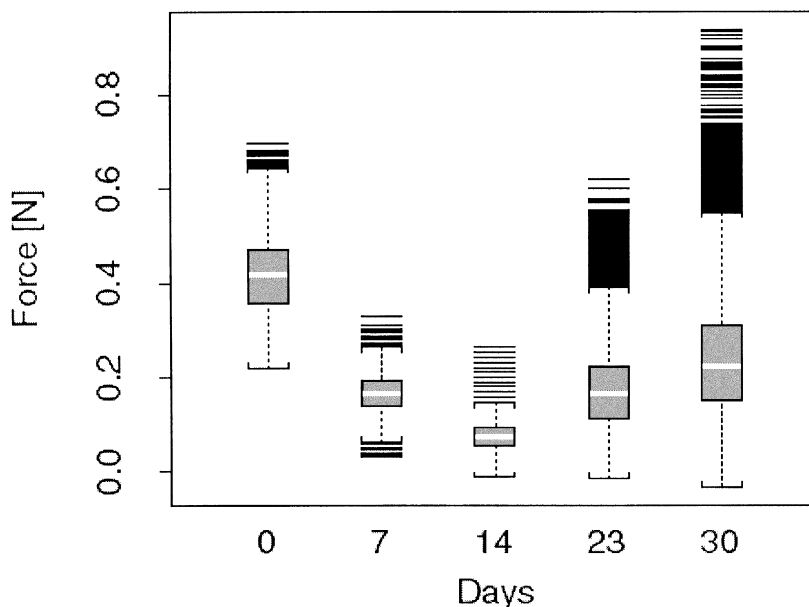


Fig. 9. Box plots of penetration resistance measurements in the TG-controlled laboratory experiment. The individual boxes show: the median as horizontal line at the interior of the box; the box itself represents the interquartile distance (IQD) (50% of data); the whiskers extend to $1.5 \times \text{IQD}$ from the center. Extreme values are shown by individual horizontal lines (most obvious at day 30).

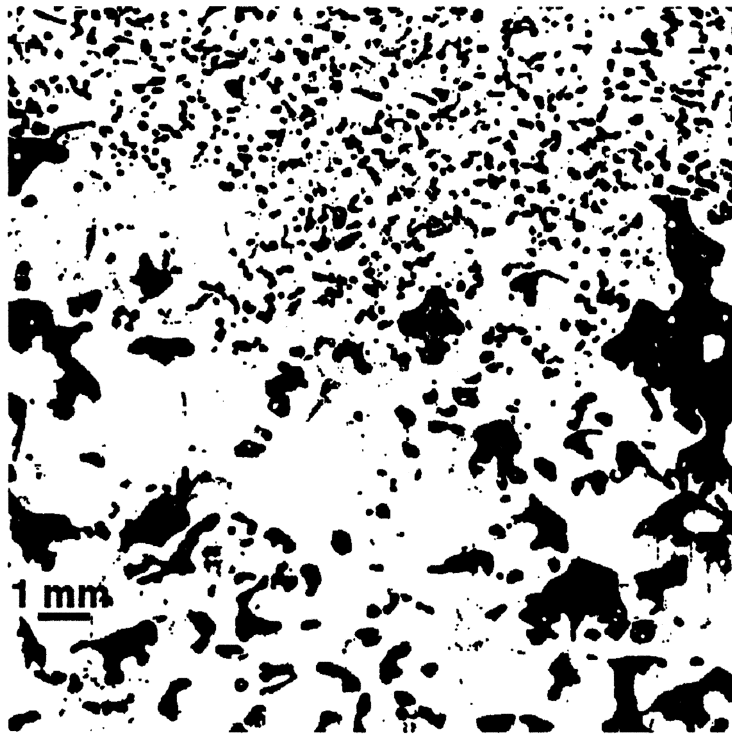


Fig. 10. Surface section of interface between melt-freeze crust and fine grained wind transported snow. Note the large pores at the interface between coarse and fine snow.

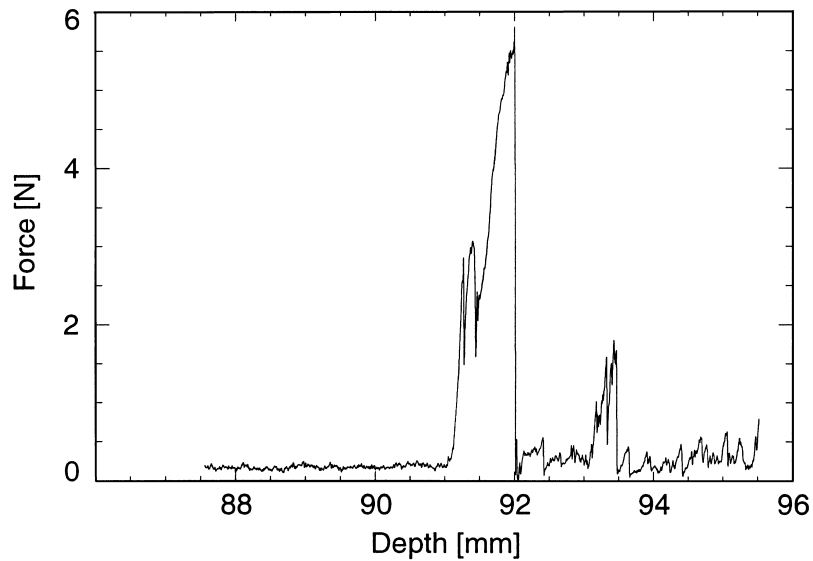


Fig. 11. Penetrometer measurement of the slab example. Fracture occurred at 95 mm.

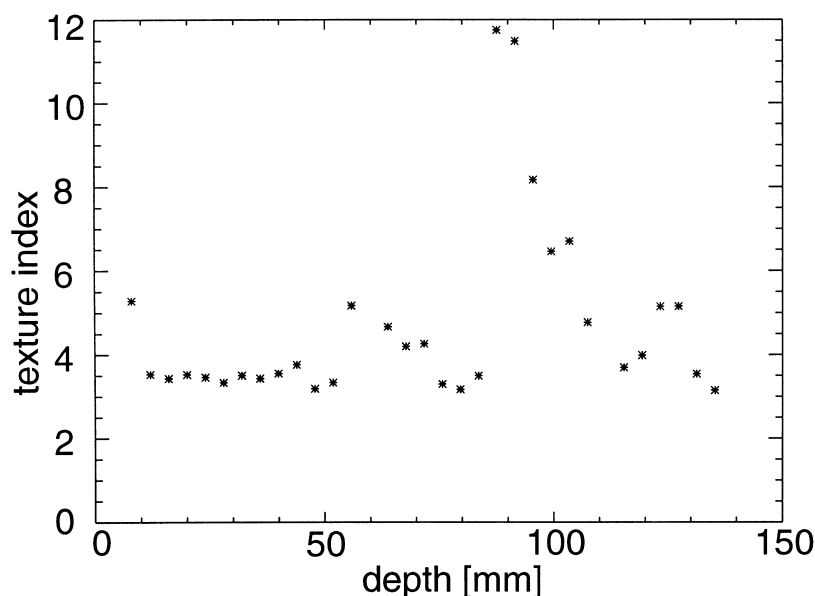


Fig. 12. The TI of slab example. The TI shows the highest value at the location of the fracture.

nal bonds are dissolved. After that an increase in the median penetration force can be observed, together with a very strong increase in the peak–valley variation of the force signal.

We observed that at after day 14 chains of grains with up to 4 and more crystals bonded together have been developed. Although strength seemed to increase, a small disturbance caused a catastrophic collapse of smaller samples. We conclude that during the formation of depth hoar two different processes occur: first a dissolution of bonds followed by a formation of bonds in the direction of the TG. Bradley et al. (1977) observed that subhedral snow (faceted grains) was the weakest stage during TG metamorphism (Fig. 8c) and not fully developed depth hoar (euhedral snow) (Fig. 8e). These measurements confirm the observations by Bradley et al. (1977).

To investigate the possibility to detect critical layers for slab avalanche formation penetration resistance was measured on two typical layer interfaces. The first interface was an about 30-cm thick layer consisting of wind-transported new snow atop a melt–freeze layer. The fracture occurred most probably at the interface between the melt–freeze layer and the fine-grained snow. Samples were taken at the crown. The interface between the melt–freeze layer

and the slab shows large holes bridging the separations between the melted grains (Fig. 10). We suspect that these holes act as a “weak layer”. The penetration measurement is shown in Fig. 11 and the calculated TI in Fig. 12.

A more classical weak layer was found at the test site of SLF at Weissfluhjoch. The profile at the relevant position has been described by T. Stucki (pers. comm.), see Table 3. The weak layer is described as partly consisting of surface hoar and faceted crystals. The position of the layers corre-

Table 3

Relevant portion of profile at test site Weissfluhjoch on 15 April 1998. The weak layer is captured *italics*. Designation of snow type follows international classification (Colbeck et al., 1990)

Height [cm] (depth [mm])	Snow type	Diameter	Hardness
174	2 (3)	0.25–1.5	2
171	2 3	0.25–0.75	1–2
169	2 (3)	0.25–1	2
167	2 (3)	0.25–1	1–2
166 (327)	3 (2)	0.25–0.5	3
<i>162 (340)</i>	<i>4 7</i>	<i>0.5–1.5</i>	<i>1</i>
<i>161 (348)</i>	<i>9e</i>	<i>0.5</i>	<i>4</i>
160 (359)	2 3	0.25–1.5	2–3
157 (392)	3 (2)	0.25–0.5	4

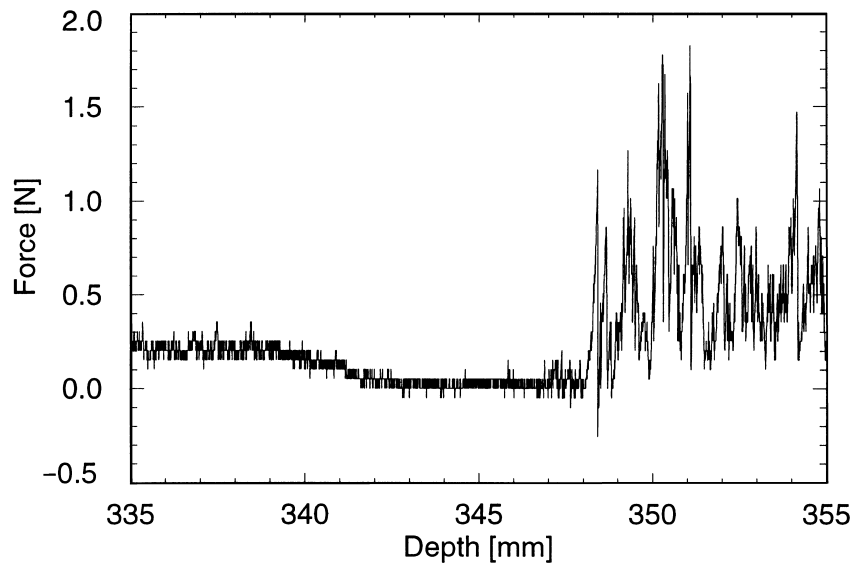


Fig. 13. Penetration measurement at the test site with weak layer between 340 and 348 mm above a melt–freeze crust.

sponds between the field profile and penetration measurement. The corresponding penetrometer measurement is shown in Fig. 13. It clearly shows between 342 and 347 mm a dip of very low resis-

tance, very similar to a signal in air, but distinctly different due to the few small spikes, at the location of the observed weak layer. The rather coarse resolution of the force signal is caused by the larger range

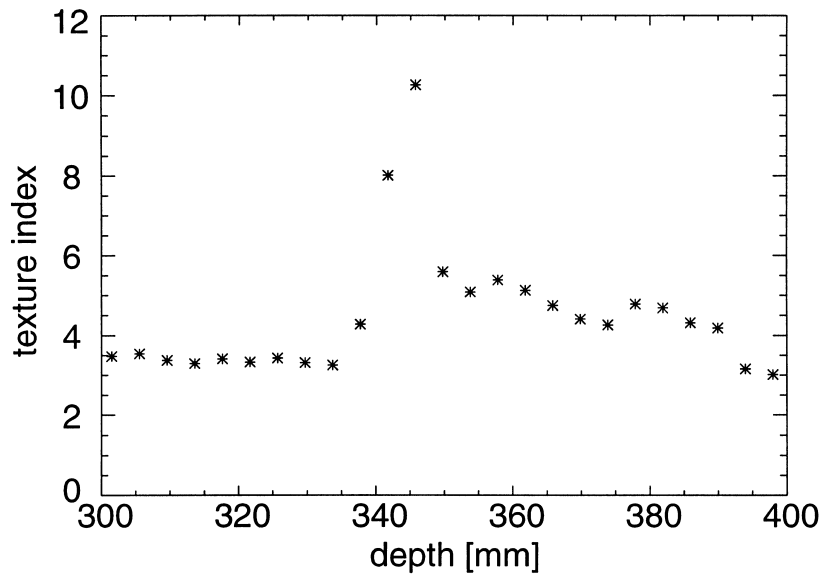


Fig. 14. The TI of the penetration measurement at the test site. The maxima is located at the position of the weak layer.

which was necessary to measure hard layers deeper in the snowpack. The weak layer is followed by a crust with a 10-fold peak resistance compared to the weak layer. The measured height of the weak layer measured with the penetrometer was larger than determined by the hand profile. This may be caused by a spatially variable thickness or the difficulty the investigate such a thin layer by hand. The TI shows the position of the critical layer by increasing from the low value of about 4–8 (Fig. 14). The change is caused by the rapid change of the penetration resistance at 348 mm. This causes an increase in the standard deviation and therefore an increase in the CV, from which TI is calculated. The TI is at the moment only well defined within a homogenous layer of snow, and not at the interface between layers.

6. Conclusions

The SnowMicroPen is able to measure the structural and spatially highly resolved mechanical parameters with a simple penetration measurement. Laboratory tests on snow metamorphosed under controlled conditions show a mechanical behavior which has only incompletely described until now because of the lack of an instrument capable of measuring such delicate materials. We could also show that the properties of interfaces between weak and hard snow layers can be measured with high precision. The SnowMicroPen makes a quantitative measure of penetration resistance and by a simple calculation also of the texture of homogenous layers, a task which in this precision would take hours instead of minutes with other methods. More work is needed to develop a classification of layers and to deduce the texture at interfaces. Even more exciting will be the link to micromechanical models (Johnson and Schneebeli,

1998). We may then be able to give an even better explanation of the mechanical behavior of a snow pack. Much more work is still needed to improve the interpretation of what constitutes a layer. The SnowMicroPen shows promise to become a significant improvement in the objective determination of layers.

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